

WINGS OF HOPE OUTREACH

Website Architecture

Complete Technical & Design Documentation

"Inspiring Hope, Transforming Lives"

Version: 1.0

Document Date: January 26, 2026

Organization: Wings of Hope Outreach Limited

Registration No: 80034655477699

Executive Summary

This document provides the complete technical architecture for the Wings of Hope Outreach website—a modern, accessible, Vercel-deployable web application featuring professional visualizations, interactive D3.js components, GSAP animations, and WCAG 2.1 AA compliant design optimized for colorblind accessibility.

The website serves as the digital presence for Wings of Hope Outreach Limited, a Uganda-based NGO incorporated on June 11, 2024, focused on supporting vulnerable children, youth, and communities through 13 comprehensive program areas.

Table of Contents

1. Organization Profile
2. Technology Stack
3. Project Structure
4. Page Architecture
5. Component Library
6. Interactive Visualizations
7. Accessibility & Colorblind Design
8. Animation System
9. Database Schema
10. API Architecture
11. Deployment Configuration
12. Design System
13. Performance Optimization
14. Security Implementation
15. Sitemap & Routing

1. Organization Profile

1.1 Legal Information

Field	Value
Legal Name	Wings of Hope Outreach Limited
Registration Number	80034655477699
Type	Company Limited By Guarantee Without Share Capital
Incorporation Date	June 11, 2024
Country	Republic of Uganda
Registered Office	Uganda

1.2 Leadership

Role	Name	Location
Director	Gabula Hannah	Wakiso, Kitala, Uganda
Secretary	Kitimbo Shellor Sarah	Wakiso, Katabitc, Kitala

1.3 Mission & Objects (13 Program Areas)

1. Basic social necessities for children and youths
2. Community-based healthcare, nutrition, water and sanitation
3. Income generating activities for socio-economic transformation
4. Partnerships with local education centers
5. Revolving micro-finance scheme for poverty eradication
6. Marriage, home and family counseling services
7. Educational institutions for academic excellence
8. Children's foster homes for vulnerable children
9. Moral guidance for children, youth, and adults

10. Financial and ethical integrity through sound governance
11. Vocational training centers
12. Religious values and ethics strengthening
13. Food crop and animal farming sensitization

2. Technology Stack

2.1 Core Framework

Layer	Technology	Version	Purpose
Framework	Next.js	14.x	React-based SSR/SSG framework
Runtime	Node.js	20.x LTS	Server runtime
Language	TypeScript	5.x	Type-safe development
Styling	Tailwind CSS	3.4.x	Utility-first CSS
Animation	GSAP	3.12.x	Professional animations
Visualization	D3.js	7.x	Data visualizations
Charts	Chart.js	4.x	Statistical charts
Maps	Leaflet	1.9.x	Interactive maps

2.2 Backend & Database

Component	Technology	Purpose
Database	PostgreSQL (Supabase)	Primary data store
ORM	Prisma	Database toolkit
Authentication	NextAuth.js	User authentication
File Storage	Cloudinary	Image/media CDN
Email	Resend	Transactional emails

6. Interactive Visualizations

The website features professional-grade interactive visualizations built with D3.js, Chart.js, and Three.js for creating engaging user experiences.

6.1 D3.js Visualization Components

- Program Wheel (Radial Dendrogram) - Interactive sunburst showing all 13 programs
- Impact Timeline - Horizontal scrollable timeline with animated entry
- Donation Flow (Sankey Diagram) - Shows donation allocation to programs
- Geographic Map (Leaflet + D3.js) - Interactive Uganda map with locations
- Program Performance Radar Chart - Multi-axis performance visualization

7. Accessibility & Colorblind Design

The website implements WCAG 2.1 AA compliance with a colorblind-safe design system to ensure maximum accessibility for all users.

7.1 WCAG 2.1 AA Compliance

- Minimum contrast ratio of 4.5:1 for normal text
- Focus indicators with 3px outline offset
- Respect for reduced motion preferences
- Minimum font size of 16px with 1.5 line height
- Screen reader compatible ARIA labels
- Keyboard navigation for all interactive elements

7.2 Colorblind-Safe Color Palette

Color	Hex Code	Usage
Teal (Primary)	#0D9488	Primary brand, buttons, links
Amber (Secondary)	#D97706	Accents, CTAs, warnings
Violet	#7C3AED	Charts, special elements
Emerald	#059669	Success states, positive metrics
Red	#DC2626	Error states, alerts
Sky	#0284C7	Information, secondary links
Pink	#DB2777	Charts, visual variety
Lime	#65A30D	Charts, progress indicators

11. Deployment Configuration

11.1 Vercel Deployment

The website is configured for Vercel Edge deployment with automatic CI/CD from GitHub, ensuring fast global delivery and seamless updates.

11.2 Estimated Resources

Resource	Specification
Database	Supabase Free Tier (upgradable)
Hosting	Vercel Pro (\$20/month)
CDN/Media	Cloudinary Free Tier (25GB)
Email	Resend Free Tier (3000/month)
Domain	wingsofhopeoutreach.org (~\$15/year)
SSL	Included with Vercel

16. Summary & Next Steps

16.1 Key Features Summary

Category	Features
Framework	Next.js 14 with App Router, TypeScript, Tailwind CSS
Visualizations	D3.js, Chart.js, Leaflet maps, Three.js
Animations	GSAP ScrollTrigger, CSS micro-interactions
Accessibility	WCAG 2.1 AA, colorblind-safe palette
Database	PostgreSQL (Supabase) with Prisma ORM
Deployment	Vercel Edge Network, automatic CI/CD
Security	CSP headers, rate limiting, Zod validation

16.2 Development Phases

Phase	Duration	Deliverables
Phase 1	2 weeks	Core structure, layout, navigation
Phase 2	3 weeks	All program pages, content management
Phase 3	2 weeks	D3.js visualizations, impact dashboard
Phase 4	2 weeks	GSAP animations, Three.js hero
Phase 5	1 week	Testing, accessibility audit
Phase 6	1 week	Deployment, domain setup, go-live

Appendix: Contact Information

Wings of Hope Outreach Limited

Registration No: 80034655477699

Wakiso, Kitala, Uganda

Director: Gabula Hannah

Secretary: Kitimbo Shellor Sarah

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